

Quantum ESPRESSO Installation Guide

- 1) `sudo apt-get update`
- 2) `sudo apt-get upgrade -y`
- 3) `sudo apt-get install -y git`
- 4) `sudo apt install --no-install-recommends \`
`autoconf \`
`build-essential \`
`ca-certificates \`
`gfortran \`
`libblas3 \`
`libc6 \`
`libfftw3-dev \`
`libgcc-s1 \`
`liblapack-dev \`
`wget`
- 5) `apt search libelpa`
(Check which version of libelpa is available for your Ubuntu version.)
- 6) `sudo apt install --no-install-recommends \`
`libopenmpi-dev \`
`libscalapack-openmpi-dev \`
`libelpa19`
- 7) `wget https://gitlab.com/QEF/q-e/-/archive/qe-7.3/q-e-qe-7.3.tar.gz`
- 8) `tar -xvzf q-e-qe-7.3.tar.gz`
- 9) `cd q-e-qe-7.3/`
- 10) `./configure`

For parallel compilation:
`./configure MPIF90=mpif90 F90=gfortran CC=mpicc`
- 11) `make all`
- 12) `cd bin`
- 13) `pwd`

Example output:
`/home/om/q-e-qe-7.3/bin`

14) nano ~/.bashrc

15) Add the following line at the end of the file:

```
export PATH=$PATH:/home/om/q-e-qe-7.3/bin
```

Press Ctrl+X to exit.

Press Y to save changes.

Press Enter to confirm.

16) source ~/.bashrc

Reloads the .bashrc file and applies the changes.

Note:

```
echo 'export PATH=$PATH:/home/om/q-e-qe-7.3/bin' >> ~/.bashrc
source ~/.bashrc
```